

INDIAN BIRD CLASSIFICATION

CNN PROJECT



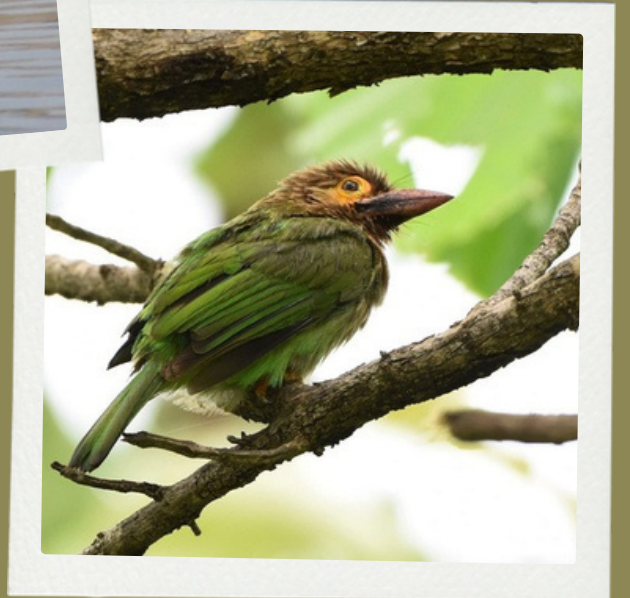
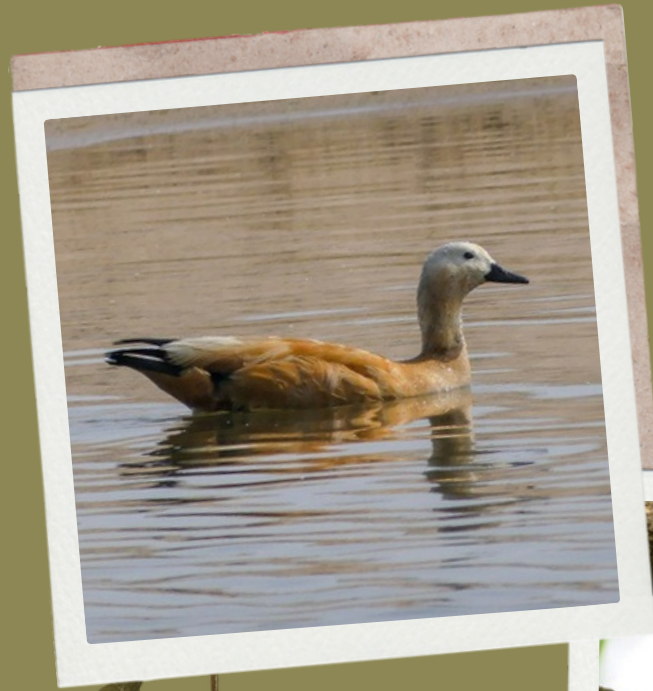
PURPOSE OF THE PROJECT

The purpose of our project is to use Convolutional Neural Networks (CNN), one of the deep learning methods, to classify bird species in India. India is a country known for its rich natural life and diversity, which leads to the existence of many different bird species. Our project aims to provide a practical tool for nature observers and enthusiasts, beyond just being a scientific study. The developed model will help users identify bird species in India and increase awareness about the conservation of wildlife.



BIRD SPECIES

"Hoopoe", "Cattle Egret", "Gray Wagtail", "House Crow", "White-Breasted Kingfisher", "Common Rosefinch", "Sarus Crane", "Northern Lapwing", "Coppersmith Barbet", "Forest Wagtail", "White Wagtail", "Brown-Headed Barbet", "Ruddy Shelduck", "Asian Green Bee-Eater", "White-Breasted Waterhen", "Common Tailorbird", "Indian Grey Hornbill", "Indian Roller", "Rufous Treepie", "Common Myna", "Jungle Babbler", "Indian Pitta", "Red-Wattled Lapwing", "Indian Peacock", "Common Kingfisher"



WHAT WE DID IN THE PROJECT?

We classified the bird species in India according to their classes. There were 25 different bird species. There were high quality photos for each bird species. Each image was labeled with the relevant bird species.

We applied various pre-processing steps to make the dataset suitable for training. We reduced all the images to a certain size (32x32 pixels). We scaled the pixel values to a certain range (255.0)

INDIAN PITTA



Did you know?
Indian Pitta can sing quite loudly.

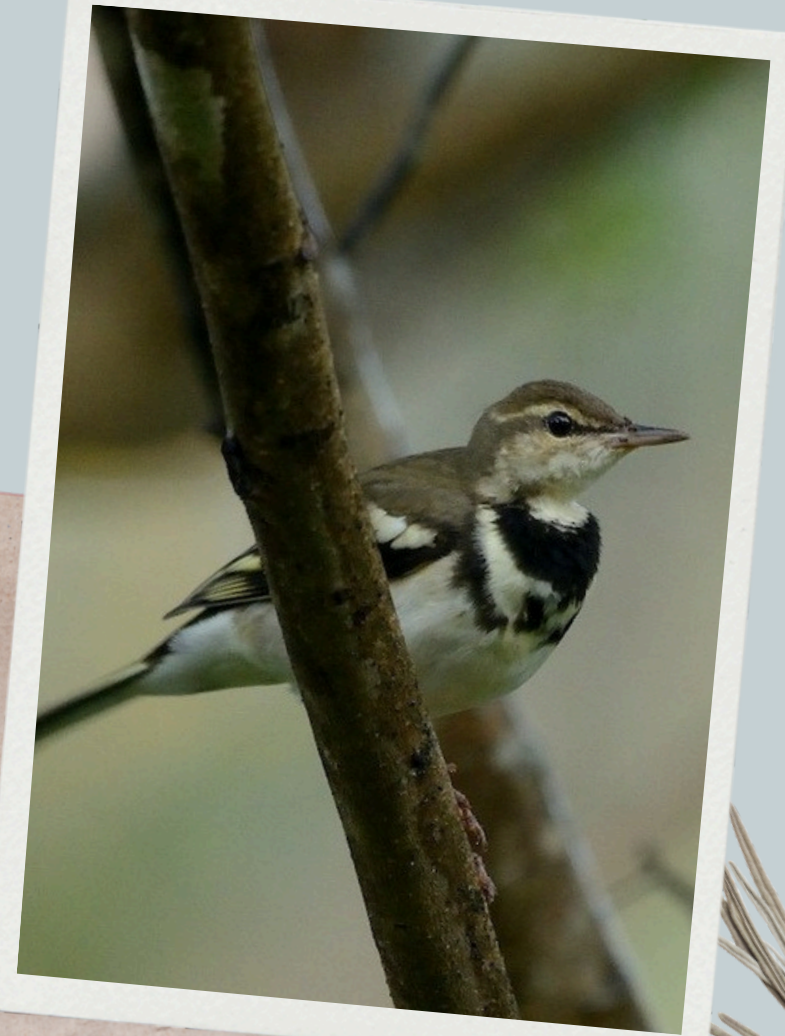
MODELLING

The model used in our project is a Convolutional Neural Network (CNN) based architecture. CNNs are specifically designed to process image data.

We took normalized images of a certain size (32x32 pixels) as input. We used a series of convolution layers. We used activation functions like ReLU. We added max pooling layers. We gave 25 neurons to the last layer since we have 25 classes. We used the softmax function in the last layer.

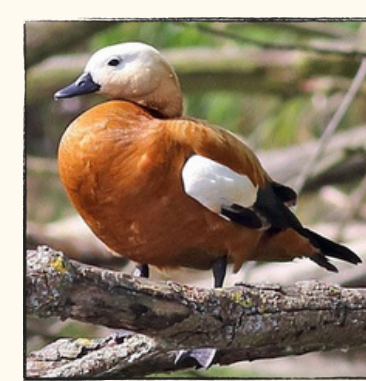
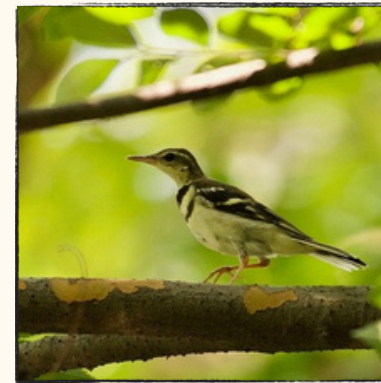


We divided the dataset into 80% training and 20% testing. The training set is used to learn the model, while the testing set is used to evaluate the overall performance of the model. During the training process, the loss function is monitored. Here we used sparse categorical crossentropy.



TRANSFER LEARNING

VGG16 is a popular deep learning architecture, often used in image classification tasks. We scaled the data in the Indian bird data and divided it into training and test data. We used the layers of VGG16. We determined our epoch number. We trained our model.



CREATING A WEBSITE

We developed a website using Streamlit with the model we saved. We uploaded it to Hugging Face. So other people can access our site and we finished our project...



NORTHERN LAPWIG



**BROWN HEADED
BARBET**



Thank you.